

Bayesian multimodeling: Bayesian inference

MIPT

2026

What do we actually want to estimate?

Suppose we observe some data $\mathbf{X}$.

What can we say about the process that generated it?

- a single “best” parameter value?
- a probability distribution over possible parameters?
- a probability distribution over future observations?

A simple example

Coin problem

A person flips a coin N times.

What is the probability of getting tails?

A simple example

Coin problem

A person flips a coin N times.

What is the probability of getting tails?

What if $N = 3$ and all three outcomes are tails?

Random variable

Given:

- a set of elementary events Ω ;
- a sigma-algebra $\mathfrak{F}$ over Ω ;
- a probability measure P .

A real-valued function

$$w : \Omega \rightarrow \mathbb{R}$$

is called a random variable if for every Borel set $B \in \mathfrak{B}(\mathbb{R})$,

$$\{\omega : w(\omega) \in B\} \in \mathfrak{F}.$$

In practice: a random variable assigns a numerical value to every outcome.

Random variable characteristics

A discrete random variable w takes a countable set of values

$$A = \{a_1, a_2, \dots\}$$

with probabilities

$$p_i = P(w = a_i), \quad \sum_i p_i = 1.$$

For a continuous random variable, we can describe its distribution using the cumulative distribution function (CDF)

$$F_w(t) = P(w \leq t),$$

or, when a density exists, the probability density function (PDF)

$$P(a \leq w \leq b) = \int_a^b f(w) dw.$$

Maximum likelihood

$$\mathbf{X} \sim f(\mathbf{x}, w),$$

$$L(\mathbf{X}, w) = \prod_{x \in \mathbf{X}} f(x, w),$$

$$\hat{w} \equiv \arg \max_w L(\mathbf{X}, w).$$

We can also use a logarithmic form:

$$\log L(\mathbf{X}, w) = \sum_{x \in \mathbf{X}} \log f(x, w),$$

$$\hat{w} \equiv \arg \max_w \log L(\mathbf{X}, w).$$

Maximum likelihood variations

Score function:

$$S(w) \equiv \frac{\partial}{\partial w} \log L(w)$$

Maximum likelihood estimator is a solution of the score equation:

$$S(w) = 0$$

Observed Fisher information:

$$I(w) \equiv -\frac{\partial^2}{\partial w^2} \log L(w)$$

MLE dispersion:

$$\mathbb{D}\hat{\theta} \approx I^{-1}(\hat{w})$$

MLE properties

- Consistency:

$$\hat{w}_n \rightarrow^p w$$

- Asymptotic normality: $n \rightarrow \infty$

$$\hat{w} \sim \mathcal{N}(w, I^{-1}(w))$$

- Efficiency: MLE's dispersion is asymptotically minimal over all the unbiased estimations
- Invariance: $g(\hat{w})$ is MLE over $g(w)$

Likelihood maximization and KL divergence

Suppose the true data distribution is $p^*(x)$ and our model is $p(x | w)$.
Then

$$KL(p^*(x) || p(x | w)) = \mathbb{E}_{x \sim p^*} \left[\log \frac{p^*(x)}{p(x | w)} \right].$$

Therefore

$$KL(p^* || p(\cdot | w)) = \text{Const} - \mathbb{E}_{x \sim p^*} \log p(x | w).$$

By the law of large numbers,

$$\mathbb{E}_{x \sim p^*} \log p(x | w) \approx \frac{1}{N} \sum_{i=1}^N \log p(x_i | w).$$

Hence, asymptotically,

$\text{MLE} \equiv \text{KL minimization}$

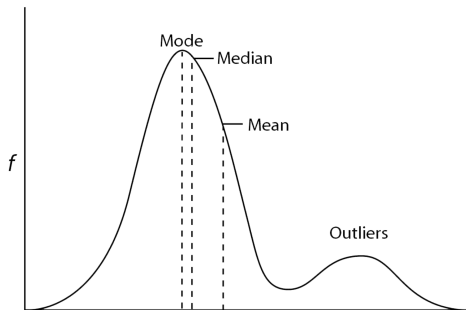
under the usual regularity assumptions.

Central tendency

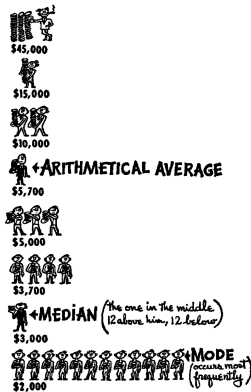
There are several reasonable ways to describe the “center” of a distribution.

- empirical mean;
- median;
- mode.

They do not have to coincide.



How statistics can be presented



Huff, 1954.

Different summaries of the same data can tell very different stories.

Quantiles and median

An α -quantile is a value w_α such that

$$P(w \leq w_\alpha) \geq \alpha, \quad P(w \geq w_\alpha) \geq 1 - \alpha.$$

Equivalently,

$$w_\alpha = F^{-1}(\alpha) = \inf\{w : F(w) \geq \alpha\}.$$

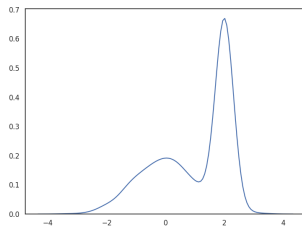
The median is the 0.5-quantile.

Mode

For a continuous distribution with density f ,

$$\text{mode}(w) = \arg \max_w f(w).$$

A distribution can have multiple modes.



Expectation

The expectation of a random variable w is

$$\mathbb{E}w = \int w dF(w).$$

Important properties:

- it may not exist;
- linear:

$$\mathbb{E}(aX + bY) = a\mathbb{E}X + b\mathbb{E}Y;$$

- changing the variable on a set of probability zero does not change the expectation;
- Law of large numbers:

$$\bar{w}_n \xrightarrow[n \rightarrow \infty]{a.s.} \mathbb{E}w;$$

- Central Limit Theorem:

$$\sqrt{n} \frac{\bar{w}_n - \mathbb{E}w}{\sqrt{Dw}} \rightarrow \mathcal{N}(0, 1).$$

Expectation: convergence

Often we have a sequence

$$w_n \xrightarrow{d/p/a.s.} w^*$$

and want to understand what happens to

$$Ef(w_n).$$

The answer depends on the type of convergence and on f .

- f is continuous and limited, then $Ef(w_n) \rightarrow^d Ef(w^*)$.
- f is continuous almost everywhere, then $f(w_n) \xrightarrow{d/p/a.s.} f(w^*)$ (Mann–Wald theorem).
- A.s. convergence: Lebesgue theorem (swap limit and expectation).

Other moments

- Variance:

$$\text{Var}(w) = \text{E}(w - \text{E}w)^2.$$

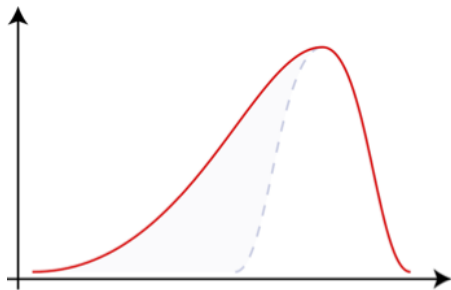
- Skewness:

$$\frac{\text{E}(w - \text{E}w)^3}{\text{Var}(w)^{3/2}}.$$

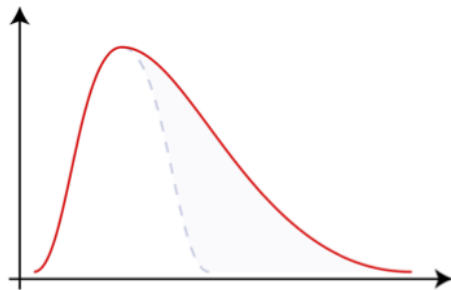
- Excess:

$$\frac{\text{E}(w - \text{E}w)^4}{\text{Var}(w)^2} - 3.$$

Skewness

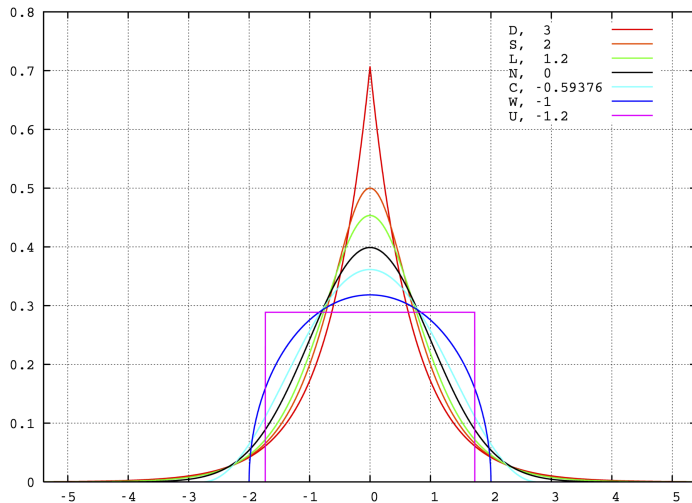


Negative Skew



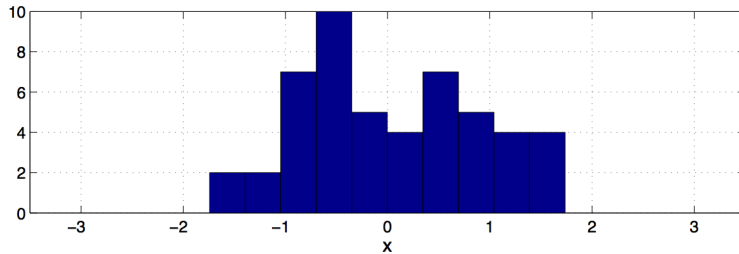
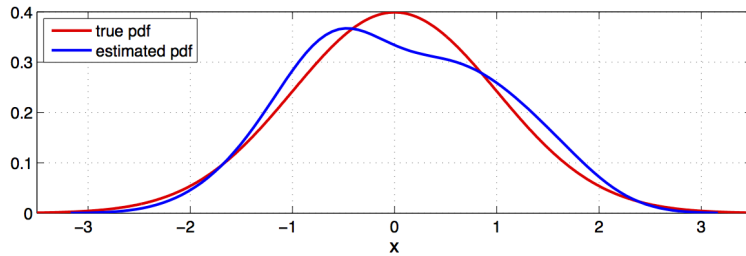
Positive Skew

Kurtosis



The fourth standardized moment describes the weight of the tails relative to a Gaussian distribution.

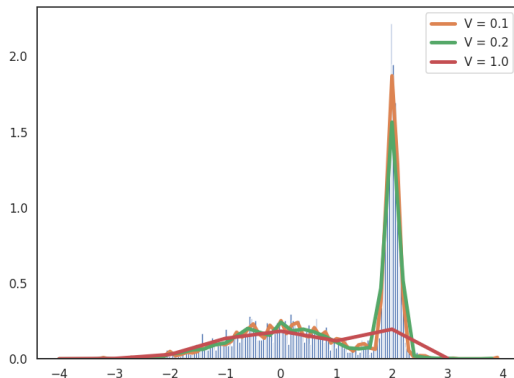
PDF estimation



KDE

Estimation using kernel function K :

$$p(x) = \frac{1}{N} \sum_{x_i \in \mathbf{X}} \frac{1}{h} K\left(\frac{x - x_i}{h}\right)$$

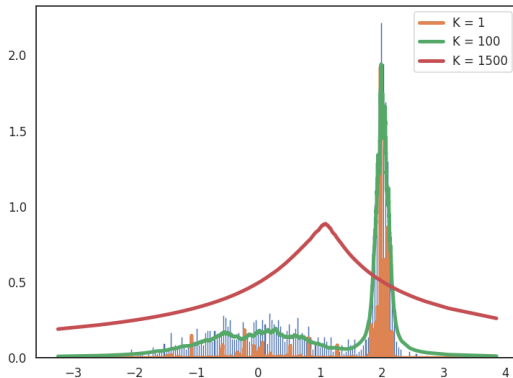


KDE

KNN can be considered as a generalization of KDE:

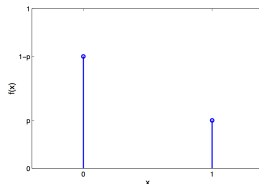
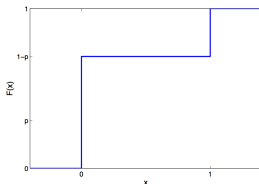
$$p(x) = \frac{1}{N} \sum_{x_i \in \mathbf{X}} \frac{1}{h} K\left(\frac{x - x_i}{h}\right),$$

$$h_i = \text{dist}(x_i, \text{neighbour}(x_i, K))$$



Bernoulli distribution

$w \in \{0, 1\} \sim \text{Ber}(p)$, $p \in (0, 1)$

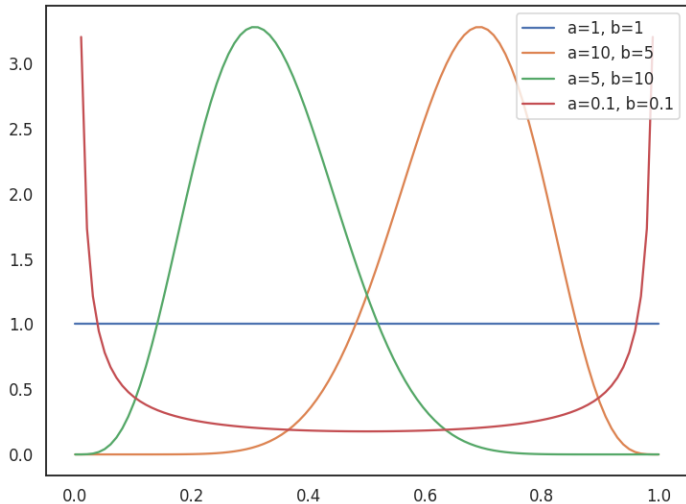


$$F(w) = \begin{cases} 0, & w < 0, \\ 1-p, & 0 \leq w < 1, \\ 1, & w \geq 1. \end{cases}$$
$$f(w) = \begin{cases} 1-p, & w = 0, \\ p, & w = 1. \end{cases}$$

- example: coin flipping

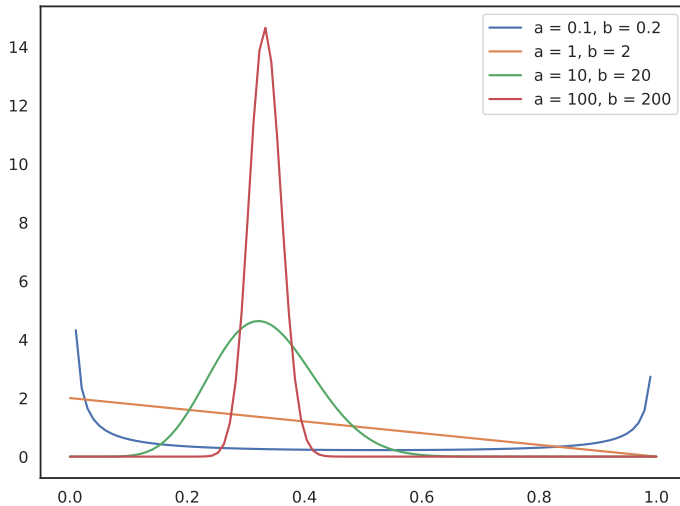
Beta-distribution

$$p(w) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} w^{a-1} (1-w)^{b-1}$$



Beta-distribution

$$p(w) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} w^{a-1} (1-w)^{b-1}$$



Beta-distribution

- corresponds to the *prior* beliefs about Bernoulli distribution
- interpretation: “effective number of events $w = 1, w = 0$ ”
- With $n \rightarrow \infty$ converges to δ -distribution with PDF concentration at MLE for Bernoulli.

Multinomial distribution

- Generalization of Bernoulli distribution for a larger number of events:

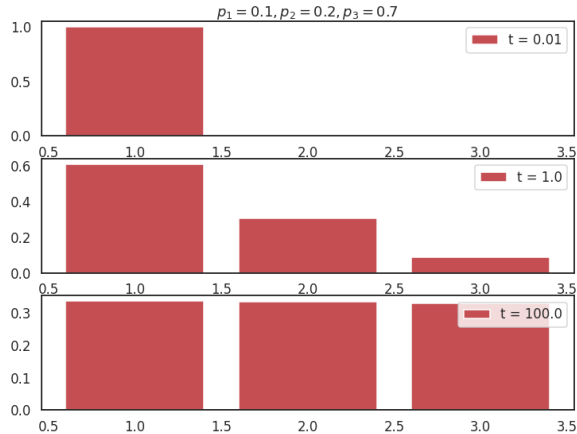
$$p(w = w_i | p_1, \dots, p_n) = p_i.$$

- Can be parametrized using logits $\mathbf{z}$ and softmax:

$$p_i = \text{softmax}(\mathbf{z}, t)_i = \frac{\exp(z_i/t)}{\sum_{j=1}^n \exp(z_j/t)}$$

- ▶ t is the temperature parameter
- ▶ if $t = 1$, softmax gives the corresponding probabilities $\mathbf{p}$

Multinomial distribution



Dirichlet distribution

$$p(w_1, \dots, w_K, \boldsymbol{\alpha}) = \frac{1}{B(\boldsymbol{\alpha})} \prod_{i=1}^K w_i^{\alpha_i - 1}$$

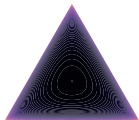
- Generalization of Beta-distribution for a larger number of events
- Support: $K - 1$ -simplex:

$$w_i \geq 0, \sum w_i = 1.$$

- Interpretation: a probability of each of K mutually exclusive events equals to w_i given that each event was observed $\alpha_i - 1$ times
- Can be parametrized using the following expression:

$$\boldsymbol{\alpha} = \bar{\boldsymbol{\alpha}} \cdot t, ||\bar{\boldsymbol{\alpha}}|| = 1.$$

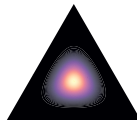
Dirichlet distribution



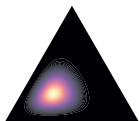
$$\bar{\alpha} = [1, 1, 1], t = 0.9$$



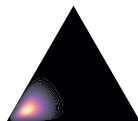
$$t = 1.0$$



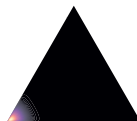
$$t = 10.0$$



$$\bar{\alpha} = [0.5, 0.25, 0.25], t = 30.0$$



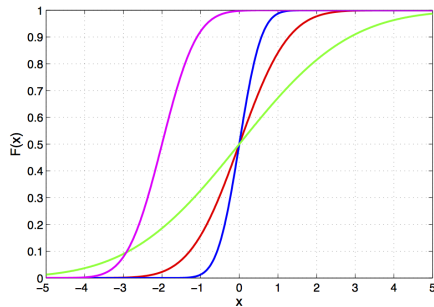
$$[0.75, 0.125, 0.125]$$



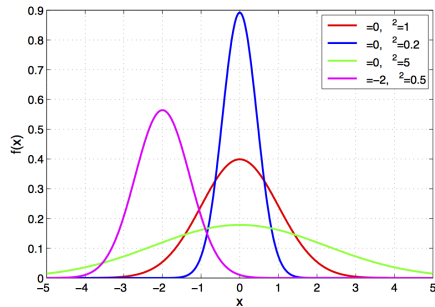
$$[0.9, 0.05, 0.05]$$

Normal distribution

$$w \in \mathbb{R} \sim \mathcal{N}(\mu, \sigma^2), \sigma^2 > 0$$



$$F(w) = \Phi\left(\frac{w - \mu}{\sigma}\right)$$
$$f(w) = \frac{1}{\sigma} \phi\left(\frac{w - \mu}{\sigma}\right)$$



$$\Phi(w) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^w e^{-\frac{t^2}{2}} dt$$
$$\phi(w) = \frac{1}{\sqrt{2\pi}} e^{-\frac{w^2}{2}}$$

Normal distribution

- limit of the sum of the weakly dependent random variables.
- $Ew = \text{median}(w) = \text{mode}(w) = \mu$, $Dw = \sigma^2$, the higher moments equal to zero
- if $w_1, \dots, w_n$ are independent, $w_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, then $\forall a_1, \dots, a_n$

$$\sum_{i=1}^n a_i w_i \sim \mathcal{N}\left(\sum_{i=1}^n a_i \mu_i, \sum_{i=1}^n a_i^2 \sigma_i^2\right)$$

- Normal distribution has the maximal differential entropy among all the continuous distributions with limited dispersion.
- example: estimation error

Student distribution

- $Ew = 0$ when $\nu > 1$, $\text{median}(w) = \text{mode}(w) = 0$ always
- let $Z \sim N(0, 1)$ and $V \sim \chi^2_\nu$ independent, then

$$T = \frac{Z}{\sqrt{V/\nu}} \sim St(\nu)$$

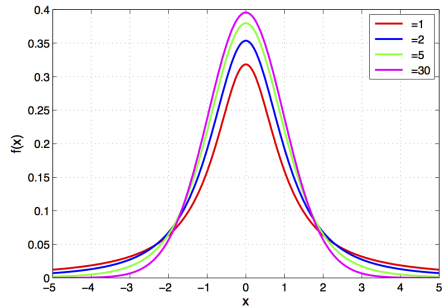
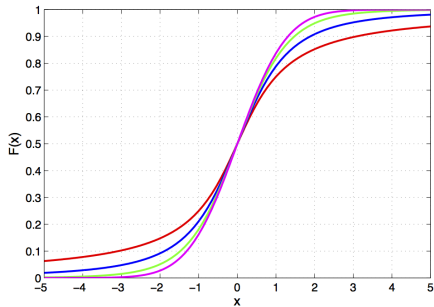
- if $w \sim St(\nu)$, then

$$Y = \lim_{\nu \rightarrow \infty} w \sim \mathcal{N}(0, 1)$$

- can be met during empirical mean estimation

Student distribution

$$X \in \mathbb{R} \sim St(\nu), \nu > 0$$



Periodical distribution

?

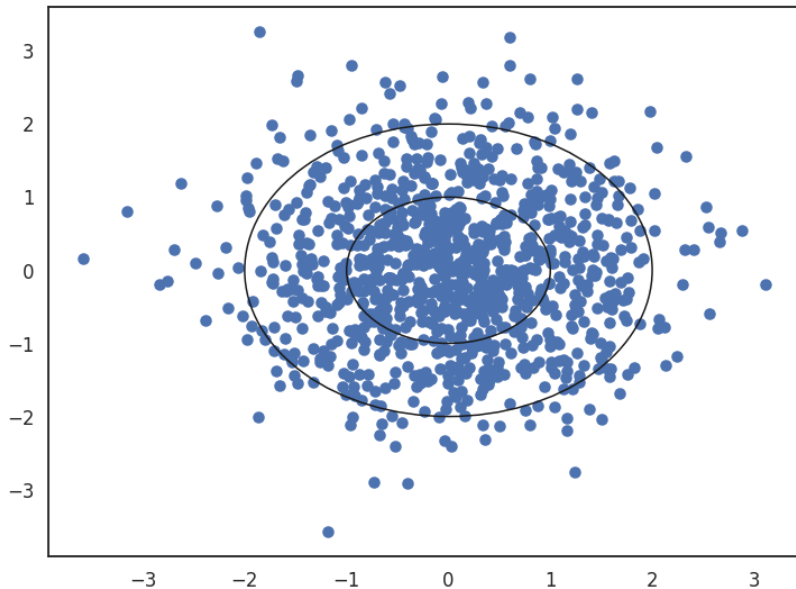
Periodical distribution

Naive approach: use a non-periodical distribution (let's say, Normal?)

Periodical distribution

- What if our dataset contains two objects on the circle: (1 degree and 359 degree)?
- $Ew = 180$, $D = 179$

Periodical distribution



Periodical distribution

- von Mises distribution:

$$p = \frac{\exp(k \cos(x - \mu))}{2\pi I_0(k)}, \quad I_0 \text{ is a Bessel function..}$$

- Distribution idea:

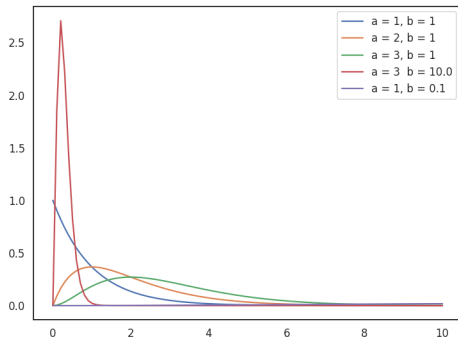
- ▶ $[w_1, w_2] \sim \mathcal{N}(\mu, \sigma^2)$
- ▶ Parameterize using polar coordinate system, limit the radius:

$$w_1 = r \cos \phi, w_2 = r \sin \phi, r = \text{Const.}$$

- See also Anton Bishuk's 2021 task 2 about Kent distribution.

Gamma distribution

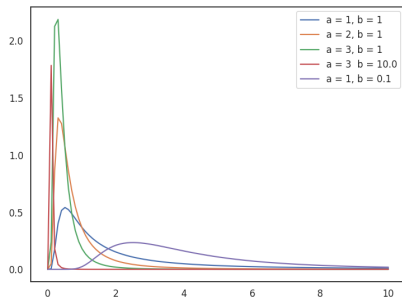
$$p(w, \alpha, \beta) = \frac{\beta^\alpha w^{\alpha-1} e^{-\beta w}}{\Gamma(\alpha)}$$



- Can be used for prior distribution modeling, for the values inverse to standard deviation.

Inverse Gamma-distribution

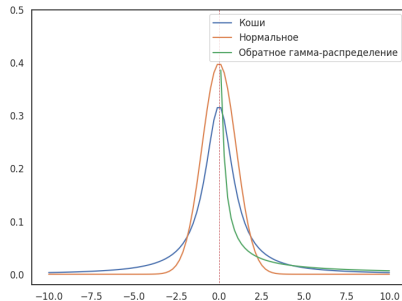
$$p(w, \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} w^{-\alpha-1} \exp(-\beta \cdot w^{-1}).$$



- $w \sim \Gamma(\alpha, \beta) \rightarrow w^{-1} \sim \text{Inv} - \Gamma(\alpha, \beta)$
- Can be used for prior distribution modeling, for the standard deviation for example.

Cauchy distribution

$$p(w, w_0, \gamma) = \frac{\gamma}{\pi ((w - w_0)^2 + \gamma^2)}.$$



- Alternative to Normal distribution and Inverse Gamma-distribution
- Heavy tails
- The moments are undetermined

Coin problem

A person flips a coin N times.
What is the probability of getting tails?

Let

$$X_i \in \{0, 1\}, \quad X_i \sim \text{Bernoulli}(w),$$

where w is the probability of tails.

Coin problem

A person flips a coin three times.
All three times it comes up tails.
What is the probability of getting tails?

We observe

$$\mathbf{X} = [1, 1, 1].$$

What should we answer?

Naive frequentist approach

The likelihood is

$$L(w; \mathbf{X}) = \prod_i w^{x_i} (1 - w)^{1-x_i}.$$

For

$$\mathbf{X} = [1, 1, 1],$$

we obtain

$$L(w) = w^3.$$

Therefore

$$\hat{w}_{\text{MLE}} = \arg \max_w w^3 = 1.$$

Is $w = 1$ really what we should believe after only three observations?

Frequentist and Bayesian statistics

Frequentist statistics

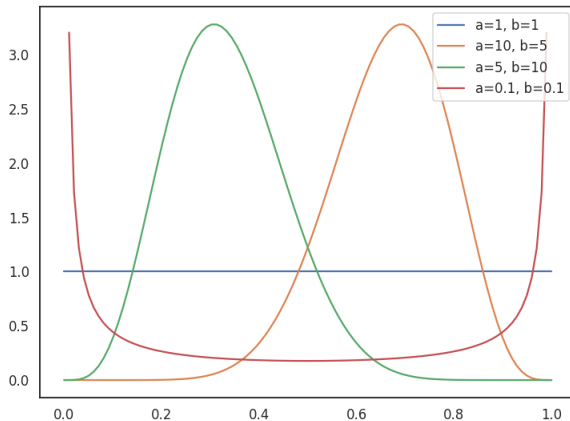
- Model parameter is a constant that is required to be estimated
- Probability is estimated purely from event frequency

Bayesian statistics

- Model parameter is a random value
 - ▶ We cannot “estimate” random value
 - ▶ But we can estimate its distribution parameters
- Probability is estimated w.r.t. our prior beliefs about data and parameter distribution
 - ▶ The more data we get the closer our estimation to MLE
 - ▶ In general our estimation is strongly relies on the prior

Beta-distribution: recap

- corresponds to the *prior* beliefs about Bernoulli distribution
- interpretation of parameters α , β : “effective number of events $w = 1, w = 0$ ”
- With $n \rightarrow \infty$ converges to δ -distribution with PDF concentration at MLE for Bernoulli.



Bayesian approach

Use beta-distribution as a *prior* distribution for our parameter w . From general considerations, the distribution should be symmetrical (unless we have more information):

$$p(w) \sim B(\alpha, \beta).$$

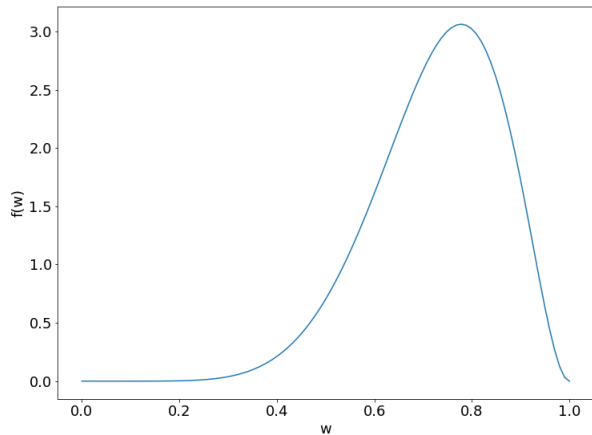
Find the *posterior* distribution of w using Bayes formula:

$$p(w|\mathbf{X}) = \frac{p(\mathbf{X}|w)p(w)}{p(\mathbf{X})} \propto p(\mathbf{X}|w)p(w);$$

$$\log p(w|\mathbf{X}) = \log p(\mathbf{X}|w) + \log p(w) + \text{Const.}$$

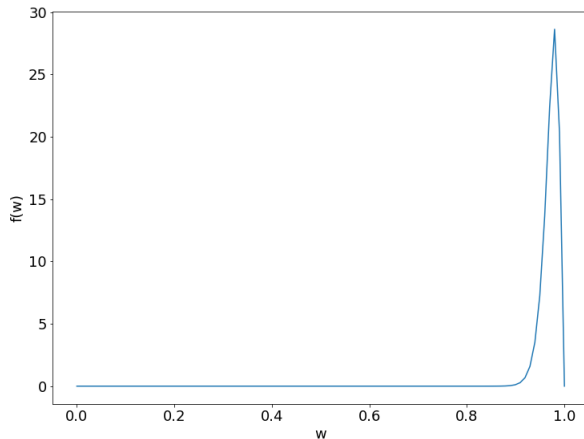
Conclusion: roughly prior is a *regularizer*.

Posterior, $\alpha = 3, \beta = 3$



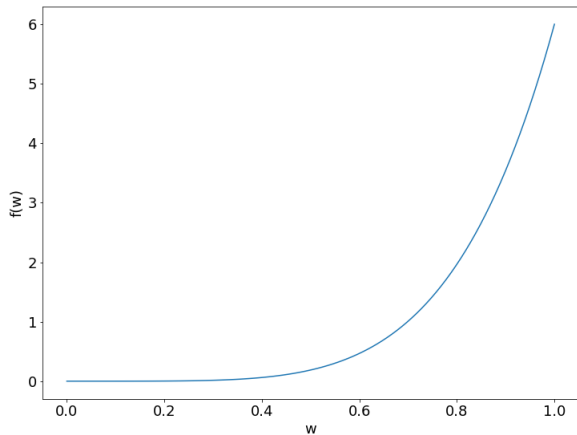
The prior can matter substantially when the amount of data is small.

Posterior after 100 observations



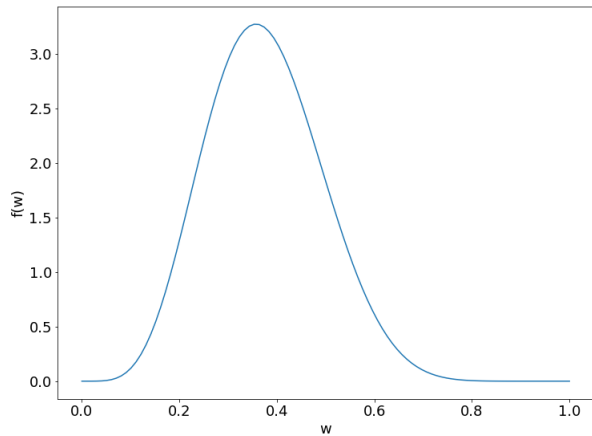
As the amount of data increases, the likelihood dominates the prior.

Posterior, $\alpha = 1, \beta = 1$



A uniform prior is possible.

Posterior, $\alpha = 1, \beta = 10$



Bayesian inference: first level

Given:

- likelihood $p(\mathbf{X}|\mathbf{w})$ of the dataset $\mathbf{X}$ w.r.t. parameters $\mathbf{w}$;
- prior distribution $p(\mathbf{w}|\mathbf{h})$
- prior parameters $\mathbf{h}$ (for the coin problem: $\mathbf{h} = [\alpha, \beta]$;))

Then the posterior for $\mathbf{w}$ w.r.t. $\mathbf{X}$:

$$p(\mathbf{w}|\mathbf{X}, \mathbf{h}) = \frac{p(\mathbf{X}|\mathbf{w})p(\mathbf{w}|\mathbf{h})}{p(\mathbf{X}|\mathbf{h})} \propto p(\mathbf{X}|\mathbf{w})p(\mathbf{w}|\mathbf{h}).$$

Find a point estimate as a maximum posterior probability (MAP):

$$\hat{\mathbf{w}} = \arg \max p(\mathbf{X}|\mathbf{w})p(\mathbf{w}|\mathbf{h}).$$

MAP-estimation is similar to MLE, if

- the dataset is large;
- prior is uniform in an infinitely large region (improper prior)

Why we used Beta-distribution?

$$\begin{aligned} p(w|\mathbf{X}, \alpha, \beta) &\propto p(\mathbf{X}|w)p(w|\alpha, \beta) \propto \\ &\propto w^{\sum x}(1-w)^{m-\sum x} \times w^{\alpha-1}(1-w)^{\beta-1} = \\ &= w^{\alpha-1+\sum x}(1-w)^{m+\beta-\sum x-1} \sim B(\alpha + \sum x, \beta + m - \sum x). \end{aligned}$$

Interpretation of parameters a, b : “effective number of events $w = 1, w = 0$ ”.

The distribution family is conjugate prior to the likelihood distribution, if the posterior belongs to the same family.

Prior families

- **Discrete (labels, discrete parameters)**

- ▶ Bernoulli
- ▶ Categorical distributions

Hyperparameters (parameters of the prior parameters):

- ▶ $w \sim \text{Bern}(w)$: $w \sim B(\alpha, \beta)$: conjugate
- ▶ $w \sim \text{Cat}(w)$: $w \sim \text{Dir}(\alpha)$: conjugate

- **Real-valued distributions**

- ▶ $\mathcal{N}$
- ▶ Laplace
- ▶ $\mathcal{C}$

Hyperparameters:

- ▶ Precision, $w \sim \mathcal{N}(\mu, \sigma^2)$, $\sigma^{-2} \in \Gamma$: conjugate for Gaussian distribution
- ▶ Expectation, $\mu \in \mathcal{N}$: conjugate for Gaussian distribution

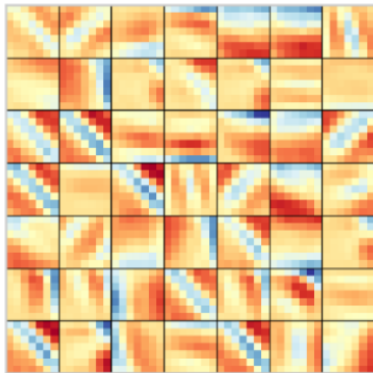
Informative prior vs Uninformative prior

- Informative prior: corresponds to some expert knowledge
 - ▶ Example: air temperature in some region: Gaussian variable with known mean and variance estimated from previous observations.
 - ▶ Mistake in informative prior estimation leads to poor models.
- Uninformative prior: corresponds to some basic knowledge
 - ▶ Example: air temperature in some region: uniform improper prior.
- Weakly-informative prior: somewhere in between
 - ▶ Example: air temperature in some region: uniform distribution in $[-50, 50]$ degrees.

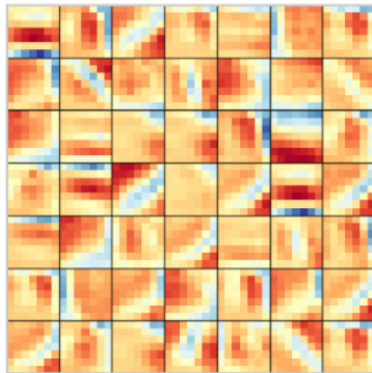
To discuss:

- $w \sim \mathcal{N}(0, \mathbf{A}^{-1})$ — what type of the prior distribution?
- What if our prior and posterior are very close

A prior can be learned



(b) Learned filters



(c) Samples from DWP

A prior does not have to be a simple analytical distribution. For example, complex models can implicitly represent distributions over neural network weights.

Jeffreys prior

Uninformative prior:

$$p(\mathbf{w}) \propto \sqrt{\det I(\mathbf{w})} = \sqrt{\det \left(-\frac{\partial^2}{\partial w^2} \log L(w) \right)}.$$

- Invariant under the variable change:

$$p(g(\mathbf{w})) = p(\mathbf{w}) \left| \frac{dg}{d\mathbf{w}} \right| \rightarrow$$

$$p(g(\mathbf{w})) \propto \sqrt{\det I(g(\mathbf{w}))}.$$

- Interpretation: a value inverse to the amount of information obtained by our model from the dataset
- Examples:
 - ▶ $y \in \text{Bin}(w) : p(w) \propto \frac{1}{\sqrt{p(1-p)}} - \text{Beta-distribution } (0.5, 0.5).$
 - ▶ $w \in \mathcal{N}(\mu, \sigma) : p(\mu) \propto \text{Const.}$
 - ▶ $w \in \mathcal{N}(\mu, \sigma) : p(\sigma) \propto \frac{1}{|\sigma|}.$

See the talk of Galina Boeva, 2023, about learning Jeffreys prior

See also talk by Daniil Dorin, 2024, about noninformative priors for small datasets

Jeffreys prior and invariance

Example:

$$X \sim \mathcal{N}(\mu, \sigma^2), \quad \sigma \text{ is known.}$$

$$p(\mu) \propto \sqrt{I(\mu)} = \text{Const} \rightarrow p(\mu) = \text{Improper prior.}$$

Change of parametrization:

$$\text{Let } \theta = \mu^3.$$

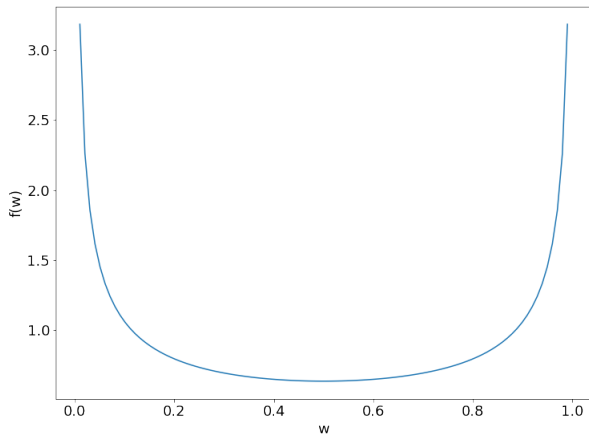
Let's use improper prior for μ and perform change of variables:

$$p(\theta) = \frac{1}{|d\theta/d\mu|} \propto \frac{1}{3\theta^{2/3}}.$$

This can be inferred by Jeffreys prior:

$$p(\theta) \propto \frac{p(\mu)}{3\theta^{2/3}} \propto \sqrt{I(\theta)}.$$

Uninformative prior $\neq$ flat prior!



See also the talk of Sergey Skorik, 2022

References

- Ширяев А. Н. Вероятность-1. – МЦНМО, 2007. – С. 552-552.
- Bishop C. M. Pattern recognition //Machine learning. – 2006. – Т. 128. – №. 9.
- KL minimization: <https://wiseodd.github.io/techblog/2017/01/26/kl-mle/>
- MacKay D. J. C., Mac Kay D. J. C. Information theory, inference and learning algorithms. – Cambridge university press, 2003.
- Atanov A. et al. The deep weight prior //7th International Conference on Learning Representations, ICLR 2019. – 2019.
- Kuznetsov M., Tokmakova A., Strijov V. Analytic and stochastic methods of structure parameter estimation //Informatica. – 2016. – Т. 27. – №. 3. – С. 607-624.
- Coin example: <https://towardsdatascience.com/visualizing-beta-distribution-7391c18031f1>
- Jefreys distribution: <https://medium.datadriveninvestor.com/firths-logistic-regression-classification-with-datasets-that-are-small-imbalanced-or-separated-49d7782a13f1>
- Atanov A. et al. The deep weight prior //arXiv preprint arXiv:1810.06943. – 2018.
- Bishop C. M., Nasrabadi N. M. Pattern recognition and machine learning. – New York : springer, 2006. – Т. 4. – №. 4. – С. 738.
- Kuznetsov M., Tokmakova A., Strijov V. Analytic and stochastic methods of structure parameter estimation //Informatica. – 2016. – Т. 27. – №. 3. – С. 607-624.
- Li Z., Chen Y., Sommer F. T. A neural network mcmc sampler that maximizes proposal entropy //arXiv preprint arXiv:2010.03587. – 2020.
- Song J., Zhao S., Ermon S. A-nice-mc: Adversarial training for mcmc //Advances in Neural Information Processing Systems. – 2017. – Т. 30.

Talks

- At least 1 talk (2 if you want to have a score ≥ 7) must be presented during the semester
- For each talk the student also provides tex files and 2 questions for Google form
- One talk must be also represented as a blog-post
 - ▶ Target audience: 2 or 3rd year student of our department
 - ▶ Language: English
 - ▶ Timing: up to 10 minutes to read
 - ▶ Focus on illustration of the approach (don't put 100 supplementary experiments from appendix)
 - ▶ No need to provide blog post ASAP: it can be done later, **BUT**...
 - ▶ ...There will be a cross-review, so keep track of time
- the topics for the next talk are on the course page (talks.md)

Project overview

Roles

Each team must assign roles for all teammates. Each role is evaluated independently, thus number of activities per each teammate must be $\approx$ equal.

- Project planning
- Proof of concept
- Algorithm implementation (**1 activity per teammate**)
- Project wrapping
- Tests writing
- Documentation writing
- Cross-review (**1 activity per teammate**)
- Final demo
- Blog post
- Tech. report

All the role must be finished according to the project schedule (see course page). For the deadline violation the score will be multiplied by 0.66 for each week.

Reparametrization trick

Reparamterization idea:

$$\varepsilon = S_{\theta}(w), \quad w = S_{\theta}^{-1}(\varepsilon).$$

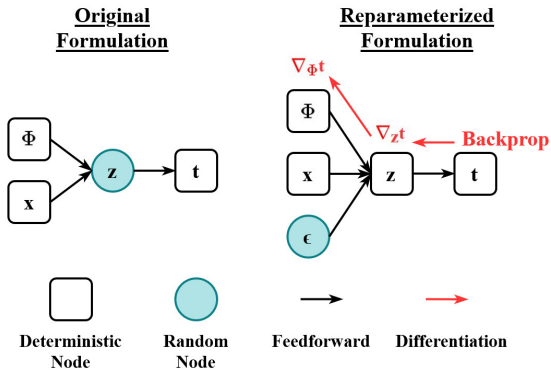
Then:

$$\nabla_{\theta} E_q f(w) = E_q \nabla_{\theta} f(S_{\theta}^{-1}(\varepsilon)).$$

Example:

$$w \sim \mathcal{N}(\mu, \sigma^2) \rightarrow S(w) = \frac{w - \mu}{\sigma} \sim \mathcal{N}(0, 1).$$

Challenge: calculation of S^{-1} is an expensive operation.



Source: wikipedia

Project: Secret energy-based models

The project proposes to consider multiple nearly-standard architectures and reveal their sampling properties: generalized (non-variational) autoencoders, classification neural networks and GAN's discriminator are proven to be energy-based models (EBM) that can be potentially used for generative tasks (sampling, out-of-domain detection, density estimation).

- Denoising autoencoder
- Classifier neural model
- GAN's discriminator

Minimum description length principle

$$\text{MDL}(\mathbf{f}, \mathcal{D}) = L(\mathbf{f}) + L(\mathcal{D}|\mathbf{f}),$$

where $\mathbf{f}$ is a model, $\mathcal{D}$ is a dataset, L is a description length in bits.

$$\text{MDL}(\mathbf{f}, \mathcal{D}) \sim L(\mathbf{f}) + L(\mathbf{w}^*|\mathbf{f}) + L(\mathcal{D}|\mathbf{w}^*, \mathbf{f}),$$

$\mathbf{w}^*$ — optimal parameters.

$\mathbf{f}_1$	$L(\mathbf{f}_1)$	$L(\mathbf{w}_1^* \mathbf{f}_1)$	$L(\mathcal{D} \mathbf{w}_1^*, \mathbf{f}_1)$
$\mathbf{f}_2$	$L(\mathbf{f}_2)$	$L(\mathbf{w}_2^* \mathbf{f}_2)$	$L(\mathcal{D} \mathbf{w}_2^*, \mathbf{f}_2)$
$\mathbf{f}_3$	$L(\mathbf{f}_3)$	$L(\mathbf{w}_3^* \mathbf{f}_3)$	$L(\mathcal{D} \mathbf{w}_3^*, \mathbf{f}_3)$

Hometask

- ① Fill the form with project preferences (will be ready soon)
- ② Kind reminder that after each classes we will have a questionnaire form with very simple and trivial questions.